

CHAPTER 1

A Preacher, a Prosecutor, a Politician, and a Scientist Walk into Your Mind



Progress is impossible without change; and those who cannot change their minds cannot change anything.

—GEORGE BERNARD SHAW

You probably don’t recognize his name, but Mike Lazaridis has had a defining impact on your life. From an early age, it was clear that Mike was something of an electronics wizard. By the time he turned four, he was building his own record player out of Legos and rubber bands. In high school, when his teachers had broken TVs, they called Mike to fix them. In his spare time, he built a computer and designed a better buzzer for high school quiz-bowl teams, which ended up paying for his first year of college. Just months before finishing his electrical engineering degree, Mike did what so many great entrepreneurs of his era would do: he dropped out of college. It was time for this son of immigrants to make his mark on the world.

Mike’s first success came when he patented a device for reading the bar codes on movie film, which was so useful in Hollywood that it won an Emmy and an Oscar for technical achievement. That was small potatoes compared to his next big invention, which made his firm the fastest-growing company on the planet. Mike’s flagship device quickly attracted a cult following, with loyal customers ranging from Bill Gates to Christina Aguilera. “It’s literally changed my life,” Oprah Winfrey gushed. “I cannot live without this.” When he arrived at the White House, President Obama refused to relinquish his to the Secret Service.

Mike Lazaridis dreamed up the idea for the BlackBerry as a wireless communication device for sending and receiving emails. As of the summer of 2009, it accounted for nearly half of the U.S. smartphone market. By 2014, its market share had plummeted to less than 1 percent.

When a company takes a nosedive like that, we can never pinpoint a single cause of its downfall, so we tend to anthropomorphize it: *BlackBerry failed to adapt*. Yet adapting to a changing environment isn’t something a company does—it’s something *people* do in the multitude of decisions they make every day. As the cofounder, president, and co-CEO, Mike was in charge of all the technical and product decisions on the BlackBerry. Although his thinking may have been the spark that ignited the smartphone revolution, his struggles with rethinking ended up sucking the oxygen out of his company and virtually extinguishing his invention. Where did he go wrong?

Most of us take pride in our knowledge and expertise, and in staying true to our beliefs and opinions. That makes sense in a stable world, where we get rewarded for having conviction in our ideas. The problem is that we live in a rapidly changing world, where we need to spend as much time rethinking as we do thinking.

Rethinking is a skill set, but it’s also a mindset. We already have many of the mental tools we need. We just have to remember to get them out of the shed and remove the rust.

SECOND THOUGHTS

With advances in access to information and technology, knowledge isn’t just increasing. It’s increasing at an increasing rate. In 2011, you consumed about five times as much information per day as you would have just a quarter century earlier. As of 1950, it took about fifty years for knowledge in medicine to double. By 1980, medical knowledge was doubling every seven years, and by 2010, it was doubling in half that time. The accelerating pace of change means that we need to question our beliefs more readily than ever before.

This is not an easy task. As we sit with our beliefs, they tend to become more extreme and more entrenched. *I’m still struggling to accept that Pluto may not be a planet*. In education, after revelations in history and revolutions in science, it often takes years for a curriculum to be updated and textbooks to be revised. Researchers have recently discovered that we need to rethink widely accepted assumptions about such subjects as Cleopatra’s roots (her father was Greek, not Egyptian, and her mother’s identity is unknown); the appearance of dinosaurs (paleontologists now think some tyrannosaurs had colorful feathers on their backs); and what’s required for sight (blind people have actually trained themselves to “see”—sound waves can activate the visual cortex and create representations in the mind’s eye, much like how echolocation helps bats navigate in the dark)*. Vintage records, classic cars, and antique clocks might be valuable collectibles, but outdated facts are mental fossils that are best abandoned.

We’re swift to recognize when other people need to think again. We question the judgment of experts whenever we seek out a second opinion on a medical diagnosis. Unfortunately, when it comes to our own knowledge and opinions, we often favor *feeling* right over *being* right. In everyday life, we make many diagnoses of our own, ranging from whom we hire to whom we marry. We need to develop the habit of forming our own second opinions.

Imagine you have a family friend who’s a financial adviser, and he recommends investing in a retirement fund that isn’t in your employer’s plan. You have another friend who’s fairly knowledgeable about investing, and he tells you that this fund is risky. What would you do?

When a man named Stephen Greenspan found himself in that situation, he decided to weigh his skeptical friend’s warning against the data available. His sister had been investing in the fund for several years, and she was pleased with the results. A number of her friends had been, too; although the returns weren’t extraordinary, they were consistently in the double digits. The financial adviser was enough of a believer that he had invested his own money in the fund. Armed with that information, Greenspan decided to go forward. He made a bold move, investing nearly a third of his retirement savings in the fund. Before long, he learned that his portfolio had grown by 25 percent.

Then he lost it all overnight when the fund collapsed. It was the Ponzi scheme managed by Bernie Madoff.

Two decades ago my colleague Phil Tetlock discovered something peculiar. As we think and talk, we often slip into the mindsets of three different professions: preachers, prosecutors, and politicians. In each of these modes, we take on a particular identity and use a distinct set of tools. We go into preacher mode when our sacred beliefs are in jeopardy: we deliver sermons to protect and promote our ideals. We enter prosecutor mode when we recognize flaws in other people’s reasoning: we marshal arguments to prove them wrong and win our case. We shift into politician mode when we’re seeking to win over an audience: we campaign and lobby for the approval of our constituents. The risk is that we become so wrapped up in preaching that we’re right, prosecuting others who are wrong, and politicking for support that we don’t bother to rethink our own views.

When Stephen Greenspan and his sister made the choice to invest with Bernie Madoff, it wasn’t because they relied on just one of those mental tools. All three modes together contributed to their ill-fated decision. When his sister told him about the money she and her friends had made, she was preaching about the merits of the fund. Her confidence led Greenspan to prosecute the friend who warned him against investing, deeming the friend guilty of “knee-jerk cynicism.” Greenspan was in politician mode when he let his desire for approval sway him toward a yes—the financial adviser was a family friend whom he liked and wanted to please.

Any of us could have fallen into those traps. Greenspan says that he should’ve known better, though, because he happens to be an expert on gullibility. When he decided to go ahead with the investment, he had almost finished writing a book on why we get duped. Looking back, he wishes he had approached the decision with a different set of tools. He might have analyzed the fund’s strategy more systematically instead of simply trusting in the results. He could have sought out more perspectives from credible sources. He would have experimented with investing smaller amounts over a longer period of time before gambling so much of his life’s savings.

That would have put him in the mode of a scientist.

A DIFFERENT PAIR OF GOGGLES

If you're a scientist by trade, rethinking is fundamental to your profession. You're paid to be constantly aware of the limits of your understanding. You're expected to doubt what you know, be curious about what you don't know, and update your views based on new data. In the past century alone, the application of scientific principles has led to dramatic progress. Biological scientists discovered penicillin. Rocket scientists sent us to the moon. Computer scientists built the internet.

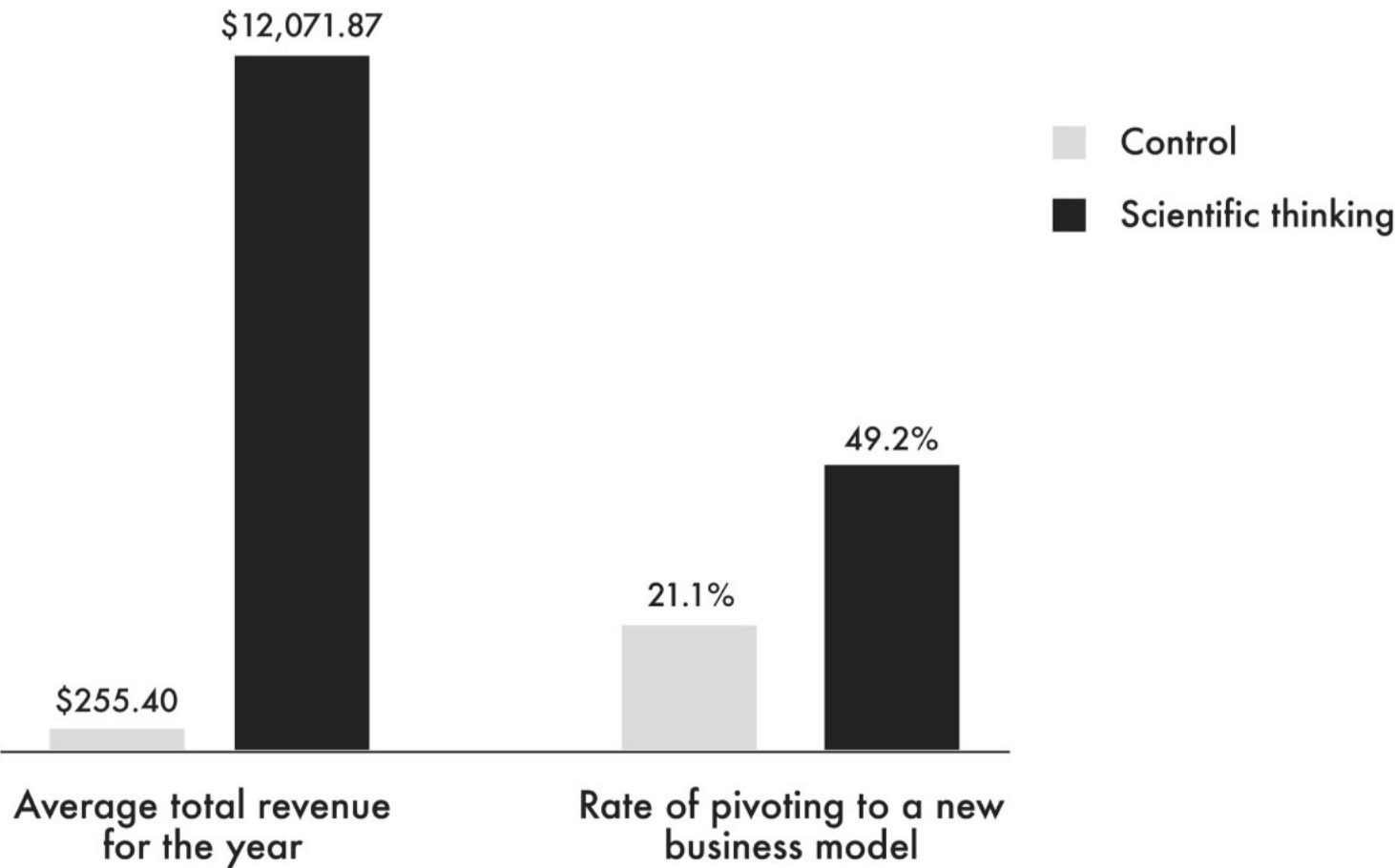
But being a scientist is not just a profession. It's a frame of mind—a mode of thinking that differs from preaching, prosecuting, and politicking. We move into scientist mode when we're searching for the truth: we run experiments to test hypotheses and discover knowledge. Scientific tools aren't reserved for people with white coats and beakers, and using them doesn't require toiling away for years with a microscope and a petri dish. Hypotheses have as much of a place in our lives as they do in the lab. Experiments can inform our daily decisions. That makes me wonder: is it possible to train people in other fields to think more like scientists, and if so, do they end up making smarter choices?

Recently, a quartet of European researchers decided to find out. They ran a bold experiment with more than a hundred founders of Italian startups in technology, retail, furniture, food, health care, leisure, and machinery. Most of the founders' businesses had yet to bring in any revenue, making it an ideal setting to investigate how teaching scientific thinking would influence the bottom line.

The entrepreneurs arrived in Milan for a training program in entrepreneurship. Over the course of four months, they learned to create a business strategy, interview customers, build a minimum viable product, and then refine a prototype. What they didn't know was that they'd been randomly assigned to either a "scientific thinking" group or a control group. The training for both groups was identical, except that one was encouraged to view startups through a scientist's goggles. From that perspective, their strategy is a theory, customer interviews help to develop hypotheses, and their minimum viable product and prototype are experiments to test those hypotheses. Their task is to rigorously measure the results and make decisions based on whether their hypotheses are supported or refuted.

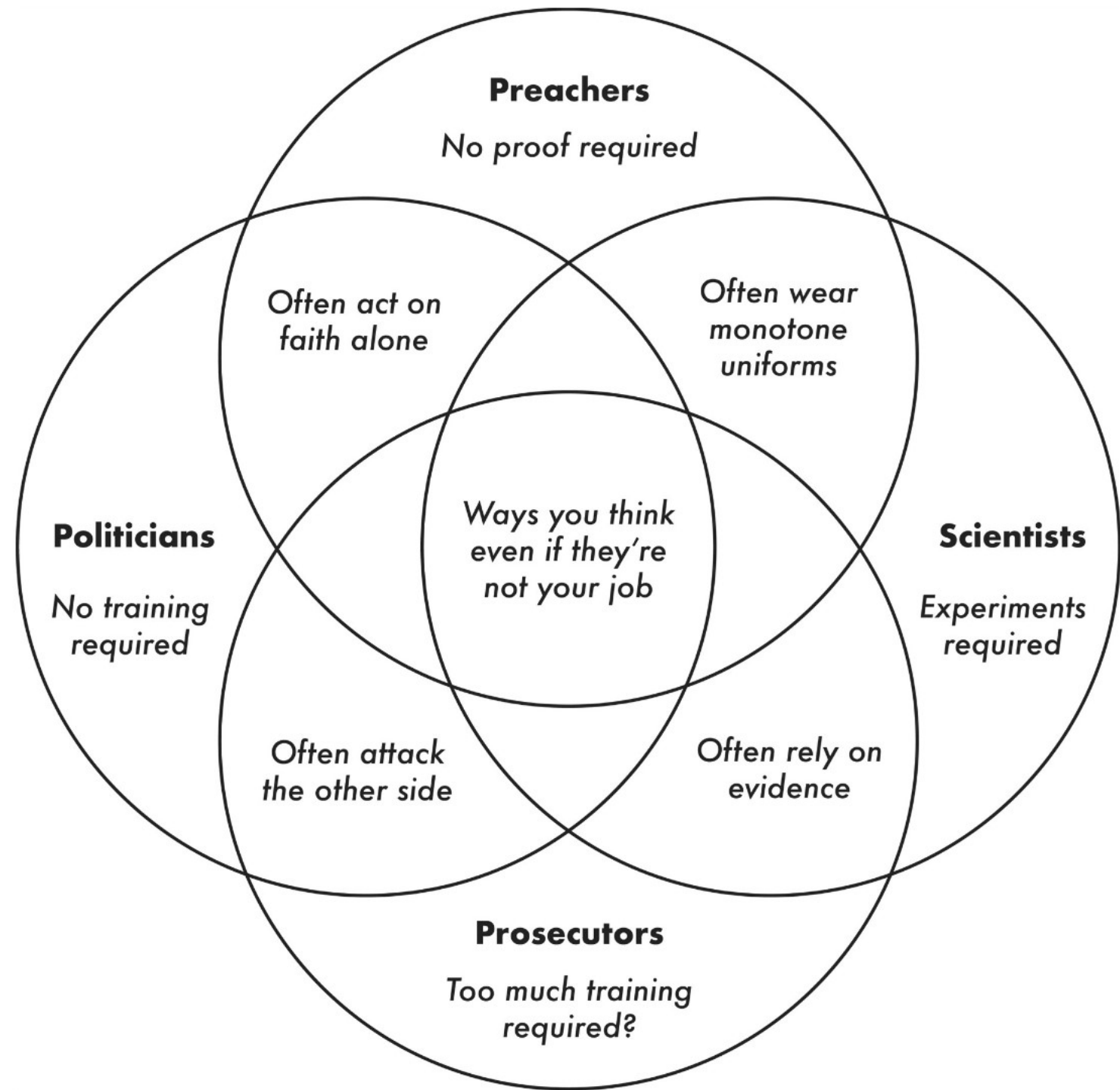
Over the following year, the startups in the control group averaged under \$300 in revenue. The startups in the scientific thinking group averaged over \$12,000 in revenue. They brought in revenue more than twice as fast—and attracted customers sooner, too. Why? The entrepreneurs in the control group tended to stay wedded to their original strategies and products. It was too easy to preach the virtues of their past decisions, prosecute the vices of alternative options, and politick by catering to advisers who favored the existing direction. The entrepreneurs who had been taught to think like scientists, in contrast, pivoted more than twice as often. When their hypotheses weren't supported, they knew it was time to rethink their business models.

THE EFFECTS OF SCIENTIFIC THINKING ON STARTUP SUCCESS



What's surprising about these results is that we typically celebrate great entrepreneurs and leaders for being strong-minded and clear-sighted. They're supposed to be paragons of conviction: decisive and certain. Yet evidence reveals that when business executives compete in tournaments to price products, the best strategists are actually slow and unsure. Like careful scientists, they take their time so they have the flexibility to change their minds. *I'm beginning to think decisiveness is overrated . . . but I reserve the right to change my mind.*

Just as you don't have to be a professional scientist to reason like one, being a professional scientist doesn't guarantee that someone will use the tools of their training. Scientists morph into preachers when they present their pet theories as gospel and treat thoughtful critiques as sacrilege. They veer into politician terrain when they allow their views to be swayed by popularity rather than accuracy. They enter prosecutor mode when they're hell-bent on debunking and discrediting rather than discovering. After upending physics with his theories of relativity, Einstein opposed the quantum revolution: "To punish me for my contempt of authority, Fate has made me an authority myself." Sometimes even great scientists need to think more like scientists.



Decades before becoming a smartphone pioneer, Mike Lazaridis was recognized as a science prodigy. In middle school, he made the local news for building a solar panel at the science fair and won an award for reading every science book in the public library. If you open his eighth-grade yearbook, you’ll see a cartoon showing Mike as a mad scientist, with bolts of lightning shooting out of his head.

When Mike created the BlackBerry, he was thinking like a scientist. Existing devices for wireless email featured a stylus that was too slow or a keyboard that was too small. People had to clunkily forward their work emails to their mobile device in-boxes, and they took forever to download. He started generating hypotheses and sent his team of engineers off to test them. What if people could hold the device in their hands and type with their thumbs rather than their fingers? What if there was a single mailbox synchronized across devices? What if messages could be relayed through a server and appear on the device only after they were decrypted?

As other companies followed BlackBerry’s lead, Mike would take their smartphones apart and study them. Nothing really impressed him until the summer of 2007, when he was stunned by the computing power inside the first iPhone. “They’ve put a Mac in this thing,” he said. What Mike did next might have been the beginning of the end for the BlackBerry. If the BlackBerry’s rise was due in large part to his

success in scientific thinking as an engineer, its demise was in many ways the result of his failure in rethinking as a CEO.

As the iPhone skyrocketed onto the scene, Mike maintained his belief in the features that had made the BlackBerry a sensation in the past. He was confident that people wanted a wireless device for work emails and calls, not an entire computer in their pocket with apps for home entertainment. As early as 1997, one of his top engineers wanted to add an internet browser, but Mike told him to focus only on email. A decade later, Mike was still certain that a powerful internet browser would drain the battery and strain the bandwidth of wireless networks. He didn’t test the alternative hypotheses.

By 2008, the company’s valuation exceeded \$70 billion, but the BlackBerry remained the company’s sole product, and it still lacked a reliable browser. In 2010, when his colleagues pitched a strategy to feature encrypted text messages, Mike was receptive but expressed concerns that allowing messages to be exchanged on competitors’ devices would render the BlackBerry obsolete. As his reservations gained traction within the firm, the company abandoned instant messaging, missing an opportunity that WhatsApp later seized for upwards of \$19 billion. As gifted as Mike was at rethinking the design of electronic devices, he wasn’t willing to rethink the market for his baby. Intelligence was no cure—it might have been more of a curse.

THE SMARTER THEY ARE, THE HARDER THEY FAIL

Mental horsepower doesn’t guarantee mental dexterity. No matter how much brainpower you have, if you lack the motivation to change your mind, you’ll miss many occasions to think again. Research reveals that the higher you score on an IQ test, the more likely you are to fall for stereotypes, because you’re faster at recognizing patterns. And recent experiments suggest that the smarter you are, the more you might struggle to update your beliefs.

One study investigated whether being a math whiz makes you better at analyzing data. The answer is yes—if you’re told the data are about something bland, like a treatment for skin rashes. But what if the exact same data are labeled as focusing on an ideological issue that activates strong emotions—like gun laws in the United States?

Being a quant jock makes you more accurate in interpreting the results—as long as they support your beliefs. Yet if the empirical pattern clashes with your ideology, math prowess is no longer an asset; it actually becomes a liability. The better you are at crunching numbers, the more spectacularly you fail at analyzing patterns that contradict your views. If they were liberals, math geniuses did worse than their peers at evaluating evidence that gun bans failed. If they were conservatives, they did worse at assessing evidence that gun bans worked.

In psychology there are at least two biases that drive this pattern. One is confirmation bias: seeing what we expect to see. The other is desirability bias: seeing what we want to see. These biases don’t just prevent us from applying our intelligence. They can actually contort our intelligence into a weapon against the truth. We find reasons to preach our faith more deeply, prosecute our case more passionately, and ride the tidal wave of our political party. The tragedy is that we’re usually unaware of the resulting flaws in our thinking.

My favorite bias is the “I’m not biased” bias, in which people believe they’re more objective than others. It turns out that smart people are more likely to fall into this trap. The brighter you are, the harder it can be to see your own limitations. Being good at thinking can make you worse at rethinking.

When we’re in scientist mode, we refuse to let our ideas become ideologies. We don’t start with answers or solutions; we lead with questions and puzzles. We don’t preach from intuition; we teach from evidence. We don’t just have healthy skepticism about other people’s arguments; we dare to disagree with our own arguments.

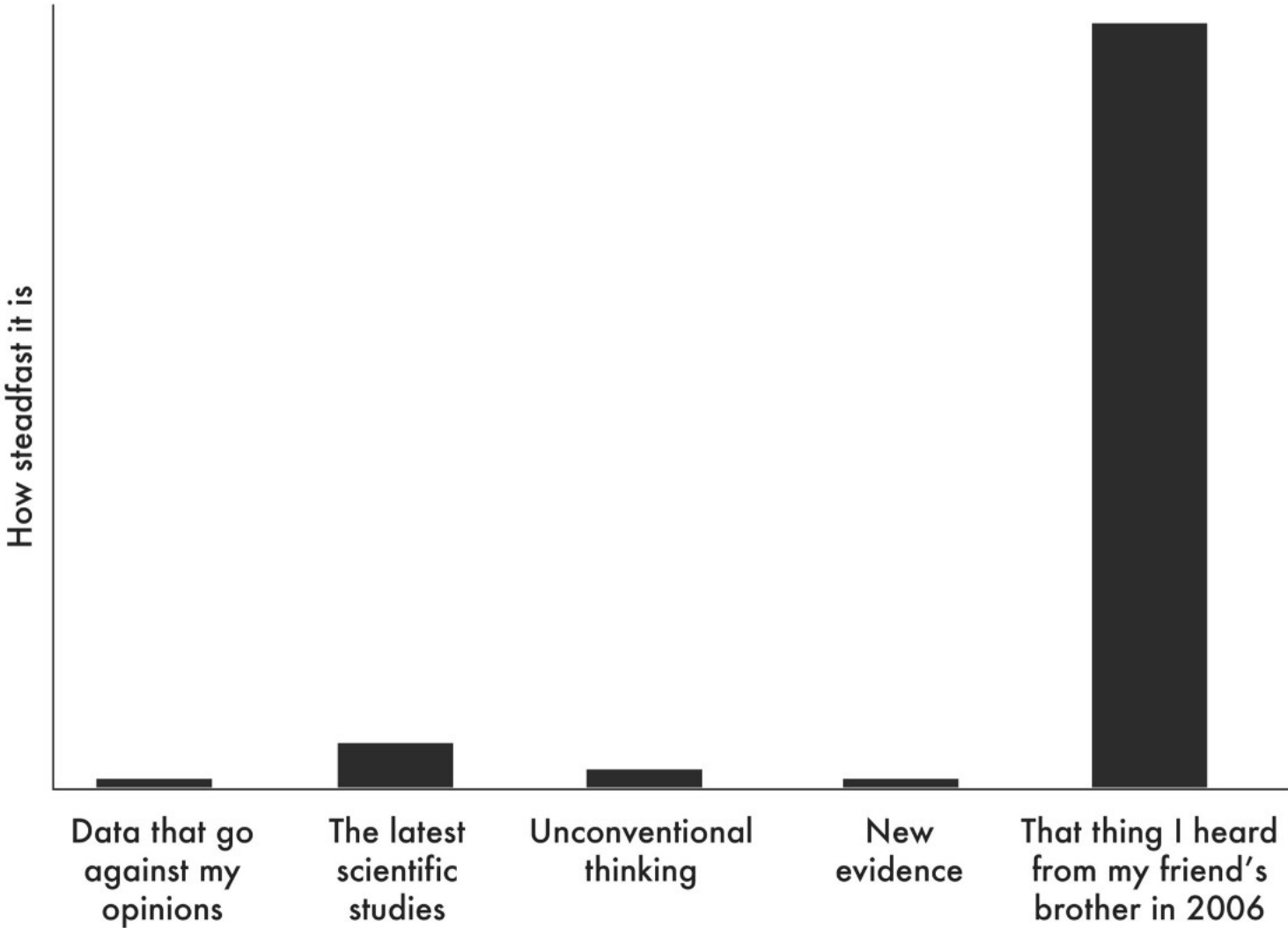
Thinking like a scientist involves more than just reacting with an open mind. It means being *actively* open-minded. It requires searching for reasons why we might be wrong—not for reasons why we must be right—and revising our views based on what we learn.

That rarely happens in the other mental modes. In preacher mode, changing our minds is a mark of moral weakness; in scientist mode, it’s a sign of intellectual integrity. In prosecutor mode, allowing ourselves to be persuaded is admitting defeat; in scientist mode, it’s a step

toward the truth. In politician mode, we flip-flop in response to carrots and sticks; in scientist mode, we shift in the face of sharper logic and stronger data.

I’ve done my best to write this book in scientist mode.* I’m a teacher, not a preacher. I can’t stand politics, and I hope a decade as a tenured professor has cured me of whatever temptation I once felt to appease my audience. Although I’ve spent more than my share of time in prosecutor mode, I’ve decided that in a courtroom I’d rather be the judge. I don’t expect you to agree with everything I think. My hope is that you’ll be intrigued by *how* I think—and that the studies, stories, and ideas covered here will lead you to do some rethinking of your own. After all, the purpose of learning isn’t to affirm our beliefs; it’s to evolve our beliefs.

BELIEFS I STAND BY



One of my beliefs is that we shouldn’t be open-minded in every circumstance. There are situations where it might make sense to preach, prosecute, and politick. That said, I think most of us would benefit from being more open more of the time, because it’s in scientist mode that we gain mental agility.

When psychologist Mihaly Csikszentmihalyi studied eminent scientists like Linus Pauling and Jonas Salk, he concluded that what differentiated them from their peers was their cognitive flexibility, their willingness “to move from one extreme to the other as the occasion requires.” The same pattern held for great artists, and in an independent study of highly creative architects.

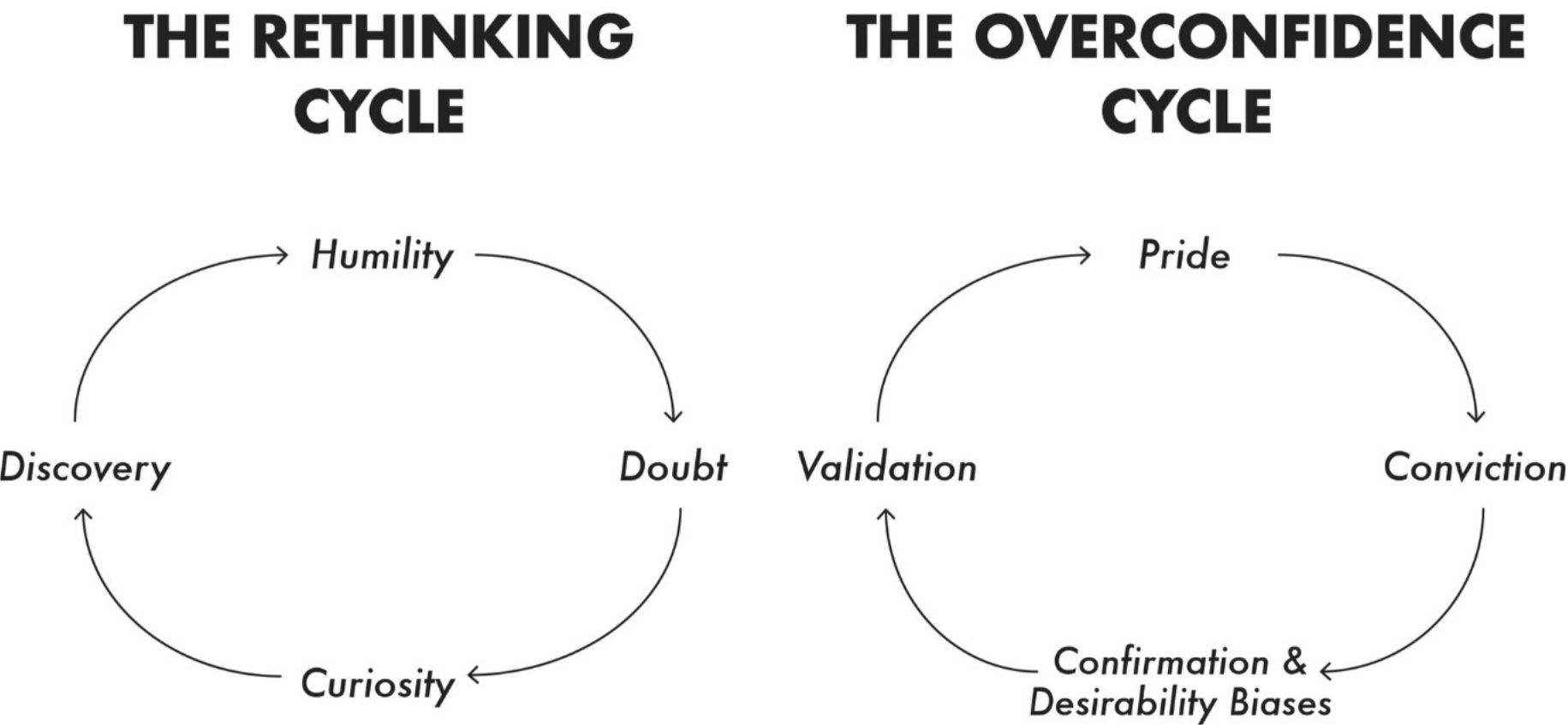
We can even see it in the Oval Office. Experts assessed American presidents on a long list of personality traits and compared them to rankings by independent historians and political scientists. Only one trait consistently predicted presidential greatness after controlling for factors like years in office, wars, and scandals. It wasn’t whether presidents were ambitious or forceful, friendly or Machiavellian; it

wasn’t whether they were attractive, witty, poised, or polished.

What set great presidents apart was their intellectual curiosity and openness. They read widely and were as eager to learn about developments in biology, philosophy, architecture, and music as in domestic and foreign affairs. They were interested in hearing new views and revising their old ones. They saw many of their policies as experiments to run, not points to score. Although they might have been politicians by profession, they often solved problems like scientists.

DON'T STOP UNBELIEVING

As I’ve studied the process of rethinking, I’ve found that it often unfolds in a cycle. It starts with intellectual humility—knowing what we don’t know. We should all be able to make a long list of areas where we’re ignorant. *Mine include art, financial markets, fashion, chemistry, food, why British accents turn American in songs, and why it's impossible to tickle yourself.* Recognizing our shortcomings opens the door to doubt. As we question our current understanding, we become curious about what information we’re missing. That search leads us to new discoveries, which in turn maintain our humility by reinforcing how much we still have to learn. If knowledge is power, knowing what we don’t know is wisdom.



Scientific thinking favors humility over pride, doubt over certainty, curiosity over closure. When we shift out of scientist mode, the rethinking cycle breaks down, giving way to an overconfidence cycle. If we’re preaching, we can’t see gaps in our knowledge: we believe we’ve already found the truth. Pride breeds conviction rather than doubt, which makes us prosecutors: we might be laser-focused on changing other people’s minds, but ours is set in stone. That launches us into confirmation bias and desirability bias. We become politicians, ignoring or dismissing whatever doesn’t win the favor of our constituents—our parents, our bosses, or the high school classmates we’re still trying to impress. We become so busy putting on a show that the truth gets relegated to a backstage seat, and the resulting validation can make us arrogant. We fall victim to the fat-cat syndrome, resting on our laurels instead of pressure-testing our beliefs.

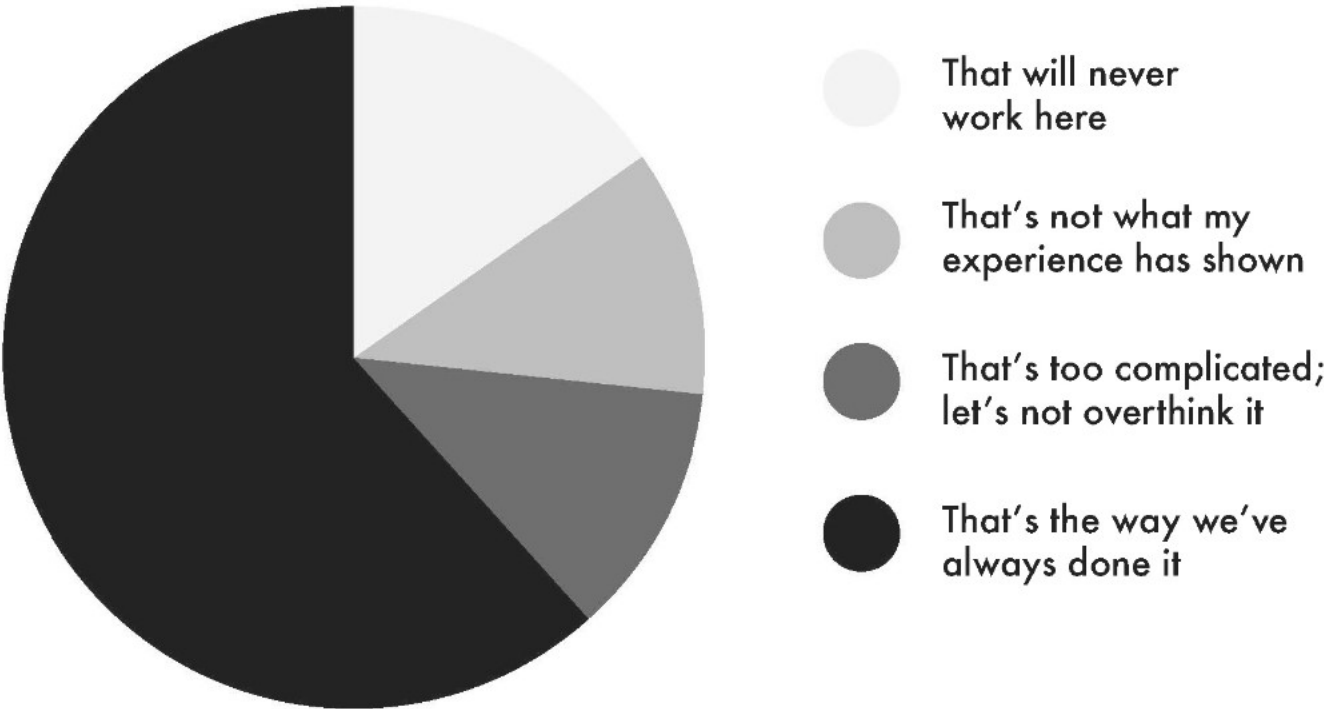
In the case of the BlackBerry, Mike Lazaridis was trapped in an overconfidence cycle. Taking pride in his successful invention gave him too much conviction. Nowhere was that clearer than in his preference for the keyboard over a touchscreen. It was a BlackBerry virtue he loved to preach—and an Apple vice he was quick to prosecute. As his company’s stock fell, Mike got caught up in confirmation bias and desirability bias, and fell victim to validation from fans. “It’s an iconic product,” he said of the BlackBerry in 2011. “It’s used by business, it’s used by leaders, it’s used by celebrities.” By 2012, the iPhone had captured a quarter of the global smartphone market, but Mike was still resisting the idea of typing on glass. “I don’t get this,” he said at a board meeting, pointing at a phone with a touchscreen. “The keyboard is one of the reasons they buy BlackBerrys.” Like a politician who campaigns only to his base, he focused on the keyboard taste of millions of existing users, neglecting the appeal of a touchscreen to billions of potential users. *For the record, I still miss the keyboard, and I’m excited that it’s been licensed for an attempted comeback.*

When Mike finally started reimagining the screen and software, some of his engineers didn’t want to abandon their past work. The failure to rethink was widespread. In 2011, an anonymous high-level employee inside the firm wrote an open letter to Mike and his co-CEO. “We laughed and said they are trying to put a computer on a phone, that it won’t work,” the letter read. “We are now 3–4 years too late.”

Our convictions can lock us in prisons of our own making. The solution is not to decelerate our thinking—it’s to accelerate our rethinking. That’s what resurrected Apple from the brink of bankruptcy to become the world’s most valuable company.

The legend of Apple’s renaissance revolves around the lone genius of Steve Jobs. It was his conviction and clarity of vision, the story goes, that gave birth to the iPhone. The reality is that he was dead-set against the mobile phone category. His employees had the vision for it, and it was their ability to change his mind that really revived Apple. Although Jobs knew how to “think different,” it was his team that did much of the rethinking.

THE MOST ANNOYING THINGS PEOPLE SAY INSTEAD OF RETHINKING



In 2004, a small group of engineers, designers, and marketers pitched Jobs on turning their hit product, the iPod, into a phone. “Why the f@*& would we want to do that?” Jobs snapped. “That is the dumbest idea I’ve ever heard.” The team had recognized that mobile phones were starting to feature the ability to play music, but Jobs was worried about cannibalizing Apple’s thriving iPod business. He hated cell-phone companies and didn’t want to design products within the constraints that carriers imposed. When his calls dropped or the software crashed, he would sometimes smash his phone to pieces in frustration. In private meetings and on public stages, he swore over and over that he would never make a phone.

Yet some of Apple’s engineers were already doing research in that area. They worked together to persuade Jobs that he didn’t know what he didn’t know and urged him to doubt his convictions. It might be possible, they argued, to build a smartphone that everyone would love using—and to get the carriers to do it Apple’s way.

Research shows that when people are resistant to change, it helps to reinforce what will stay the same. Visions for change are more compelling when they include visions of continuity. Although our strategy might evolve, our identity will endure.

The engineers who worked closely with Jobs understood that this was one of the best ways to convince him. They assured him that they weren’t trying to turn Apple into a phone company. It would remain a computer company—they were just taking their existing products and adding a phone on the side. Apple was already putting twenty thousand songs in your pocket, so why wouldn’t they put everything else in your pocket, too? They needed to rethink their technology, but they would preserve their DNA. After six months of discussion, Jobs finally became curious enough to give the effort his blessing, and two different teams were off to the races in an experiment to test whether they should add calling capabilities to the iPod or turn the Mac into a miniature tablet that doubled as a phone. Just four years after it launched, the iPhone accounted for half of Apple’s revenue.

The iPhone represented a dramatic leap in rethinking the smartphone. Since its inception, smartphone innovation has been much more incremental, with different sizes and shapes, better cameras, and longer battery life, but few fundamental changes to the purpose or user experience. Looking back, if Mike Lazaridis had been more open to rethinking his pet product, would BlackBerry and Apple have compelled each other to reimagine the smartphone multiple times by now?

The curse of knowledge is that it closes our minds to what we don’t know. Good judgment depends on having the skill—and the will—to open our minds. I’m pretty confident that in life, rethinking is an increasingly important habit. Of course, I might be wrong. If I am, I’ll be quick to think again.